

communicate and share information to create smart environments.

An image processing technique called Perspective-n-Point (PnP) is used, where an image of an object is captured and then compared with a second image, in which four LEDs in a known configuration illuminate the object. This helps the camera to estimate the position of the object within a few centimetres.

Now transform a bicycle into a smartphone-powered smart vehicle

There are many ways of using a smartphone to navigate a bike trip, but these mostly consist of checking a handset or smartwatch, both potentially dangerous activities while steering a bike.

SmartHalo provides a green-lighting prompt, connected via Bluetooth to an app on a smartphone, allowing users to keep the handset in the pocket as they follow the colour-coded navigation guides on the device. All they have to do is input the destination into the device's associated app and SmartHalo takes over from there.

When they are about to turn, the signal beams a white and green warning, and when going in the wrong direction, they get a red flashing prompt. Another lighting feature notifies them when they get a phone call. The device also connects with the smartphone app to provide biking statistics, including distance travelled, average speed and calories burned. It also includes an alarm, among other many interesting features.

Disney to launch Linux based LEDs that communicate with each other

Researchers at Disney have showcased light bulbs that can do more than just illuminate rooms. These Linux based light emitting diodes (LEDs) can communicate with each other, with objects and with the Internet, to create smart environments.

Transmitting signals via light is an age-old concept. The researchers, however, have created networking technology that makes it possible for LED lights to not only communicate with each other but do so in a way that is compatible with the Internet and its technical protocols. They used off-the-shelf commercial LED light bulbs, which they modified so that these could send and receive visible light signals.

These modifications include a system on chip running Linux operating system, visual light communication (VLC) controller module with the protocol software and an additional power supply for added electronics. They created software that makes the signals transmitted through this hardware compatible with Internet protocols. They were thus able to create networks with a throughput of up to one kilobit per second.

These VLC-enabled bulbs could be used to broadcast

beacons, making it possible to detect the location of objects, linked into a network to route signal traffic or be used to communicate with objects.

According to Stefan Mangold, head of Disney Research's wireless research group, these advances could give VLC an important role in the growing Internet of Things idea.

A robotic vacuum cleaner that can see

The new vacuum cleaner Roomba 980 from iRobot has an enhanced array of sensors that helps it clean better, using visual inputs. The robotic cleaner incorporates a low-resolution video camera that helps it map the house as it learns the layout of rooms and location of objects like furniture and household items.

It works on iRobot's proprietary visual simultaneous localisation and mapping (vSLAM) technology for the first time in a consumer product. Using the map, the device runs continuously for up to two hours, after which it automatically return to Home Base to recharge and resume until the cleaning is done. It also pairs with an app, enabling users to start a cleaning cycle remotely via the Internet.

Digitsole unveils shoe insole that counts steps and keeps toes warm

Digitsole Warm Series insoles, by Digitsole, have a thick heel that tapers dramatically from the arch forward to a flat, slightly cushioned area for the toes. The



The battery in Digitsole insoles gives seven to eight hours of continuous operation and then recharges via twin USB chargers

back area has an impact cushion for the heel, as well as a lot of the core technology inside the sole. It has motion sensors that track steps, distance and calories. The arch also includes a micro-USB charging port which connects to heating elements that run through the front half of the insole. The product is also waterproof.

It connects via Bluetooth to a dedicated iOS or Android app where the user can track his or her activity stats and adjust the sole's thermostat. It can heat up to 30°C in -25°C for at least six hours.